

Article

Evaluating the Interactive and Transformative Role of Innovation, Education, Human Capital and Natural Resources Policies in Protecting and Sustaining Environmental Sustainability

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Abstract: Achieving environmental sustainability remains a critical challenge for governments worldwide, particularly within the G20 bloc, due to rapid urbanization, resource-intensive industrial activities, and the environmental pressures associated with globalization. Despite various efforts, ecological degradation continues to escalate, necessitating a deeper understanding of the factors influencing environmental sustainability. This study investigates the role of technological innovation (TLI), education (EDU), human capital (HMC), and natural resources (NTS) in shaping ecological sustainability, while also assessing the effects of globalization (GLN) and urbanization (URZ) on the ecological footprint (EFT) from 2000 to 2022. By employing the Driscoll and Kraay standard error (DKSE) method, the study provides robust empirical insights into these relationships. The findings reveal that TLI, EDU, and HMC significantly reduce EFT, supporting the notion that innovation and human capital development contribute to environmental sustainability. Conversely, NTS, GLN, and URZ exacerbate ecological degradation, underscoring the environmental costs of resource exploitation and urban expansion. These results remain consistent when validated through the CS-ARDL robustness test. Furthermore, the Dumitrescu-Hurlin causality test identifies reverse causality between NTS, EDU, and EFT, while unidirectional causality is confirmed for the remaining variables. The study's findings highlight the necessity for policymakers to implement eco-friendly technological advancements, sustainable education frameworks, and responsible resource management strategies to mitigate environmental degradation. These insights contribute to the broader discourse on sustainable development and provide actionable recommendations for fostering long-term ecological balance in G20 nations.

Keywords: technological innovation; human capital; education; natural resources; urbanization; sustainable development goals



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1. Introduction

Human activities that need energy, water, fiber, lumber, infrastructure, and other resources put strain on the environment, resulting in loss of biodiversity and climate change. The EFT represents the fertile land and ocean needed to meet human use of NTS and absorb the waste produced by human actions [1]. The current human need for NTS has surpassed their biocapacity, leading to an overshoot scenario. This means that the use of available resources exceeds its capacity to replenish them. According to [2],

it takes almost eighteen months to replenish the resources that we use during a year. The widening disparity between EFT and biocapacity diminishes the Earth's capacity for productivity, leading to pollution, food scarcity, and biodiversity loss [3]. Consequently, individuals migrate to metropolitan regions due to enhanced employment prospects, improved access to healthcare, and better educational options. The process of economic and social modernization drives URZ. Metropolitan migration and the conversion of rural land into urban properties are the defining features of this phenomenon [4]. A significant demographic shift has occurred in the global population, with over 50% of people now living in metropolitan regions. In addition, it is estimated that by 2050, 66% of the population will live in urban regions which results in the inclusion of about 2.5 billion individuals to the urban population [5]. URZ exacerbates the population growth of cities that already have constrained resources. As a result, the requirements for food, energy, transportation, and public utilities are growing, which in turn contributes to climate change, rapid pollution, and depletion of resources [6]. URZ has a detrimental influence on local food production and contributes to emissions, despite its positive effects on economic growth, innovation, and knowledge [7]. Ref. [8] found that it leads to a reduction in soil fertility. Produces significant amounts of garbage, contributes to the expansion of deforestation, and leads to environmental deterioration [8]. Urban dwellers use about 75% of Earth's NTS, exceeding 66% of the global energy consumption, and produce around 70% of the overall greenhouse gas (GHG) emissions [9].

Despite the growing body of research on environmental issues, human activities continue to accelerate environmental degradation, including the depletion of natural resources (NTS) and the worsening effects of climate change. Addressing these challenges requires a strong emphasis on education, training, and awareness to enhance environmental quality (ENQ). Prior studies have established a link between education (EDU), environmental consciousness, human capital (HMC), and ENQ. HMC, which encompasses both educational attainment and the skills acquired through education, plays a crucial role in shaping sustainable behaviors. For instance, [10] highlights that HMC influences consumer preferences for renewable energy products, while [11] emphasize the role of education in understanding the drivers and impacts of climate change. Moreover, [12] demonstrates the positive impact of education on recycling efforts, and [13] argues that educational awareness enhances compliance with environmental laws. Additionally, [14] finds that education contributes to controlling forest degradation, while [15] shows that HMC significantly reduces CO₂ emissions by promoting energy efficiency. In the context of G20 countries, where economic growth and industrialization exert significant environmental pressure, the role of HMC becomes even more critical. Moreover, ref. [16] suggests that enhancing HMC can substantially mitigate the ecological footprint (EFT). Given the diverse economic structures and environmental policies across G20 nations, investing in education and skill development is essential for fostering sustainable consumption, green innovation, and responsible resource management. Strengthening HMC in these economies could drive long-term environmental sustainability by integrating environmental consciousness into policy frameworks and societal behavior.

Moreover, several researchers have examined the connection between URZ and environmental quality, with diverse outcomes. Refs. [4,17] contends that URZ leads to environmental deterioration via the amplification of energy use. In contrast, several academics propose that URZ mitigates environmental deterioration via the facilitation of innovation and the adoption of green technologies [18]. The results demonstrate that the effects of URZ are contingent upon the management of the urban population, the extent of URZ, and several other aspects. The empirical literature mainly focuses on CO₂ to investigate its impact on environmental quality, therefore only addressing a fraction of the ecological

issues linked to energy use [19]. Whereas this current study aims to examine the influence of TLI, HMC, NTS, URZ, GLN, and EDU on environmental quality in G20 nations.

In connection with the environmental study, the EDU has a crucial role in promoting ENQ and mitigating CO₂ emissions and EFT. EDU facilitates fostering Research and development (R&D) in the field of conversion and energy storage technologies. It also has the potential to enhance environmental consciousness, hence promoting the use of renewable energy sources [2]. Many studies have shown a correlation between EDU and a propensity to comply with environmental regulations [13], embrace green technology and goods [10], and mitigate environmental damage [20,21]. R&D in green technology is essential for harnessing the advantages of clean energy. According to Sinha et al. [22] achieving SDGs is essential for promoting technological progress and EDU. The literature shows that the industrial revolution and the growing global population have led to a higher demand for power generation, resulting in more consumption of NTS. This, in turn, has led to a rise in GHG emissions. Numerous studies, including those by [23] argue that energy demand has a significant impact on environmental quality. Energy consumption is vital for economic growth, but it often results in environmental deterioration. When economies grow, the higher energy demand harms the environment if sourced from fossil fuels; however, sustainable energy sources can significantly mitigate this impact [24].

Hence, it is imperative to prioritize environmental protection with the pursuit of SSD [25]. Following the UN-SDGs in 2015, governments have made attempts to attain these objectives. Nevertheless, progress towards the SDGs has been limited, as shown by [26]. The UN has convened several significant meetings and conferences on climate change and GLN. The Kyoto Protocol and the Paris Climate Agreement are the two most significant accords in this area. The methods implemented to address environmental deterioration are aligned with the SDGs, and they also include green technology [27]. Furthermore, the recent Conferences COP26 and COP27 have significance due to the resolutions adopted and the matters that garnered significant interest. Moreover, at the COP26 conference, countries made a promise to reduce the GHGs, stop the loss of forests, attain a state of net zero emissions by the year 2050, speed up the phase-out of coal usage, and stop world funding for fossil fuels [28]. Furthermore, the Conference COP27 highlighted the significance of shifting towards a decarbonized economy [29].

Moreover, it is evident from the literature that SSD is a worldwide occurrence that greatly affects living beings [30]. Additionally, the heavy reliance on fossil fuels is a challenge for nations in attaining SDG-7, which centers on ensuring clean energy availability [31]. Whereas, the energy production and consumption of nations have a significant impact on attaining eco-friendly economic developments [32]. Therefore, many countries face significant challenges in achieving SDG 12, which centers on promoting responsible consumption and production, as well as SDG 13, which focuses on taking action to deal with climate change. On the other hand, an important factor contributing to this challenge is the overuse of resources in the process of industrialization and GDP growth [33]. This overconsumption of NTS presents significant risks, including forest loss and climate change [34]. Although NTS are not considered environmentally toxic, their extraction poses a significant threat to the ecosystem [35].

The focus on the G20 panel is particularly relevant, as these nations represent the world's largest economies and are among the most significant contributors to environmental degradation due to their high levels of industrialization, energy consumption, and resource exploitation [36]. Recognizing their global environmental responsibility, G20 countries have actively engaged in international climate initiatives and policy reforms. Many of these nations have committed to global environmental agreements, including the Paris Agreement, established at the COP21 conference, which brought together 190 countries [37].

A key objective of this agreement is to limit global temperature rise to a maximum of 2 °C, with efforts to further constrain it to 1.5 °C. Specific commitments include reducing methane emissions by 30% by 2030, curbing deforestation linked to imported goods, phasing out coal consumption, and eliminating inefficient fossil fuel subsidies—measures that were reaffirmed at COP27 [38]. Despite these efforts, the implementation of climate policies in G20 countries remains in its early stages, largely centered around regulatory frameworks rather than comprehensive action. In this context, our study examines the relationships between technological innovation (TLI), human capital (HMC), education (EDU), natural resources (NTS), globalization (GLN), and urbanization (URZ) with ecological footprint (EFT) in G20 countries from 2000 to 2022. Utilizing the DKSE method and the CS-ARDL robustness test, we aim to provide empirical insights into how these factors influence environmental sustainability in major economies. Figure 1 illustrates the trends in EFT across G20 nations over the study period.

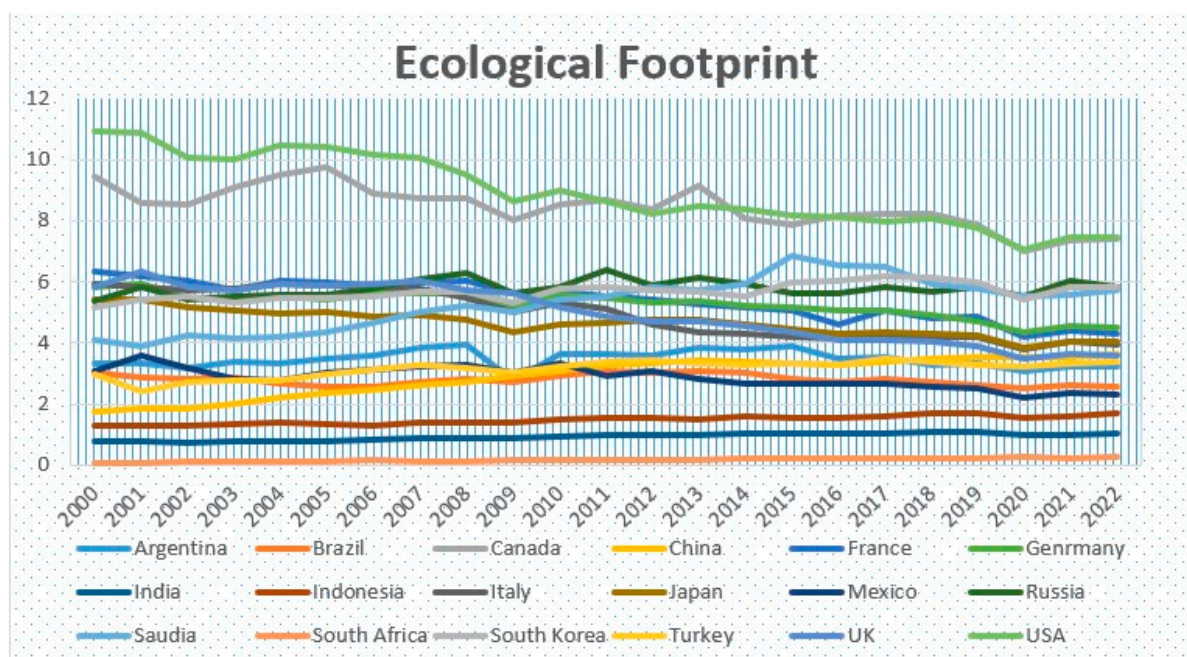


Figure 1. EFT trends in G20.

Achieving environmental sustainability remains a critical challenge for G20 nations due to their diverse economic structures, rapid urbanization, and resource-intensive industrial activities. The ecological footprint (EFT) serves as a comprehensive measure of environmental degradation, yet limited research has collectively examined the influence of technological innovation (TLI), human capital (HMC), education (EDU), natural resources (NTS), globalization (GLN), and urbanization (URZ) on EFT within the G20 framework. While previous studies have primarily focused on carbon emissions (CO₂) and greenhouse gases (GHGs), a broader perspective incorporating multiple socio-economic and technological factors is needed. Moreover, Figure 2 sums up the remaining sections of the study.

This study aims to investigate the collective impact of TLI, HMC, EDU, NTS, URZ, and GLN on the EFT of G20 nations under the framework of the Sustainable Development Goals (SDGs). It further examines the distinct roles of HMC and EDU in shaping ecological sustainability, as most studies treat HMC as a proxy for education, despite EDU being only one component of HMC. Additionally, it assesses whether URZ, GLN, and NTS contribute to environmental degradation or support sustainability efforts. Given the conflicting evidence in the literature, this study employs the Driscoll and Kraay Standard Error (DKSE) method, along with CS-ARDL for robustness checks, to ensure the reliability of findings.

The results will guide policymakers in integrating technological advancements, sustainable education policies, and resource management into environmental strategies. Based on the research objectives, this study seeks to address the following key questions:

1. How does TLI impact the environmental quality of G20 nations?
2. Do human capital and education contribute to improving the environmental quality of G20 nations?
3. What roles do globalization and urbanization play in influencing the environmental quality of G20 countries?
4. Do natural resource extraction and usage contribute to environmental deterioration in G20 countries?
5. How far are G20 countries from achieving the SDGs set by the UN?

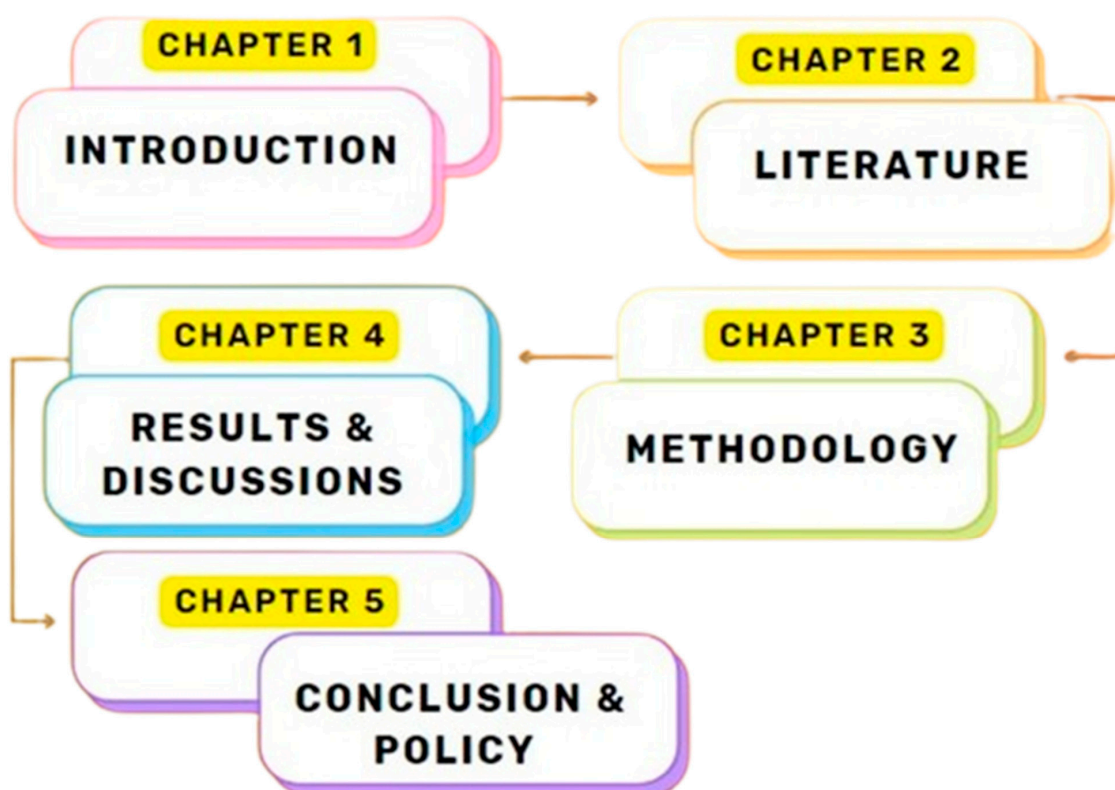


Figure 2. Remainder of Study.

2. Literature Review

EFT is studied as a comprehensive indicator that is used to assess the ecological impacts of human activities. The current studies have shown a preference for EFT over CO₂ as a measure of SSD [39,40]. In 2015, the United Nations released the 2030 Agenda, which included 17 SDGs and 169 associated targets. Therefore, it is crucial to use integrated approaches to establish and illustrate the connections between the SDGs and accelerate their successful implementation via Cleaner Production practices and concepts to fulfill the demands of the environment [41]. Studies show that there is a direct connection between ENQ and economic activities. They claim that these activities often contribute to increased carbon emissions if they do not utilize environmentally friendly technology [42]. To provide policy insights applicable to all areas, it is essential to evaluate the elements of SSD in G20 nations that exhibit comparatively superior pollution control measures. This research aims to examine the relationship between HMC, TLI, EDU, GLN, and URZ in G20 countries from 2000 to 2022.

2.1. Human Capital and Environmental Quality Nexus

The role of HMC in the sustainable transition and its impact on EFT has been relatively underexplored in historical studies [43]. However, existing research suggests that HMC significantly contributes to environmental sustainability. For instance, [20] found that highly skilled human resources help reduce EFT through energy-efficient practices, while [44] emphasized that HMC plays a vital role in managing and decreasing energy consumption. Similarly, refs. [45,46] highlighted that skilled human capital enhances manufacturing efficiency and productivity by adopting eco-friendly technologies. Several studies confirm that HMC fosters environmental awareness and promotes sustainability initiatives. Ref. [10] linked education to eco-friendly product advocacy, while [12] stressed its role in resource recovery. Additionally, HMC facilitates clean energy adoption [47] and encourages responsible innovation [48]. Furthermore, ref. [13] found that education and training significantly influence the adoption of renewable energy and energy-efficient practices.

Most studies exploring the relationship between HMC and ENQ have relied on CO₂ emissions as a key indicator of environmental quality. For instance, ref. [49] used quantile regression to confirm that educated human resources contribute to CO₂ reduction across 30 Chinese provinces. Similarly, ref. [50] employed GMM and fixed effects to demonstrate HMC's role in mitigating CO₂ emissions, supporting the EKC hypothesis. Other studies, such as [51] in China and [20] in Pakistan, confirmed the reciprocal association between HMC and environmental deterioration. Ref. [52] utilized QARDL to show that the impact of HMC on environmental sustainability varies across quantiles in Brazil and China. Ref. [53] further emphasized that HMC and ICT together reduce CO₂ emissions across 70 nations. However, findings on HMC's impact on ENQ remain mixed. While studies on G7 countries [2] indicated that HMC helps lower EFT, others, such as [54] in GCC nations, found that HMC does not necessarily reduce environmental degradation. These conflicting results highlight the need for further research on the nuanced role of HMC in different economic and environmental contexts.

2.2. Technological Innovation and Environmental Quality Nexus

Technological progress plays a pivotal role in addressing environmental challenges. While advancements in environmental technology contribute to pollution control by reducing harmful emissions, their effectiveness depends on accessibility and practical application. Some studies highlight the potential of technological innovation (TLI) in enhancing environmental quality [55]. Experts emphasize that technological advancements are essential in mitigating greenhouse gas (GHG) emissions and reversing environmental decline [56]. Empirical studies support this notion, demonstrating that TLI helps reduce CO₂ emissions in various regions, including Asian countries [57], OECD nations [58], and the European Union [59]. Similarly, [60] argue that TLI significantly contributes to CO₂ reduction, while [61] found that the impact of TLI varies across income levels, being more effective in high-income nations. Additionally, ref. [62] suggest that technological advancements enhance energy efficiency and drive the transition to renewable energy, further promoting ecological sustainability.

Emerging economies, particularly in the G20, play a crucial role in the advancement of green technology and sustainable innovation. Countries like China, India, and Brazil are investing heavily in renewable energy, digital infrastructure, and low-carbon technologies, positioning themselves as key players in global sustainability efforts [63]. The rapid industrialization of these nations presents both opportunities and challenges—on the one hand, technological diffusion accelerates green transitions, while on the other, weak regulatory frameworks and uneven access to innovation may limit effectiveness [64]. Despite significant progress, disparities in research funding, policy implementation, and

technology transfer mechanisms hinder emerging economies from fully leveraging TLI for environmental gains [65]. Addressing these gaps requires targeted policies that promote inclusive access to clean technologies, cross-border collaborations, and knowledge-sharing platforms to ensure that sustainability efforts are both scalable and impactful.

However, despite its potential benefits, technological innovation is not universally regarded as an effective means of improving environmental quality. Some studies argue that TLI may lead to unintended ecological imbalances and economic disparities, particularly if its implementation is uneven or if it prioritizes industrial expansion over sustainability [66,67]. These researchers caution that rather than alleviating environmental degradation, TLI could sometimes exacerbate ecological concerns. Thus, while technology remains a crucial tool for environmental protection, its success depends on strategic implementation that ensures accessibility and sustainability while minimizing adverse economic and ecological consequences.

2.3. Education and Environmental Quality Nexus

EDU is essential in reducing EFT and CO₂. Refs. [68,69] argue that EDU is vital for countries to achieve victory in the battle against climate change. EDU enhances energy efficiency [70]. Several studies have used EDU as a surrogate for educational attainment due to the EDU index being composed of numerous elements, with EDU being a significant component. Several researchers conducted a study on the correlation between EDU, carbon emissions, and energy consumption. They used various methodologies and found conflicting results. Ref. [70] investigated the correlation between EDU and energy consumption in OECD countries, resulting in a negative relationship between EDU and energy usage. The author proposed that EDU, which is considered one of the most vital components, leads to the attainment of energy efficiency. Ref. [71] gathered data at the industry level in India and examined the influence of the EDU index on SO₂ and NO₂ using the panel technique. Their findings show that EDU has a negative relation with SO₂ and a positive relation with NO₂. [72] conducted a study on the influence of EDU on methane and CO₂ in 181 world economies, using panel methods. The findings indicate that EDU has a negligible impact on CO₂, whereas it has a notable and adverse impact on methane emissions. Ref. [73] examined the influence of FDI and EDU on polluting emissions in Latin American nations, within the context of the EKC, while also considering additional control factors. Their panel approach findings revealed an inverse correlation between EDU and pollution in high-income nations, whereas a positive correlation was seen in low-income ones. Ref. [74] examined the relationship between EDU and ENQ in Australia, using the EKC framework. The ARDL findings elucidated the curvilinear relationship between EDU and emissions, characterized by an inverted U shape. The U-shaped pattern suggests that EDU plays a crucial role in Australia's ability to reduce emissions after a certain threshold is reached. Ref. [20] used many indicators to assess EDU levels in Pakistan and examine the correlation with CO₂ by applying the ARDL method. A negative correlation was seen between carbon emissions and EDU. Ref. [75] used a panel technique and found consistent findings for the OECD nations. The authors highlight the significance of EDU in promoting environmental sustainability. Ref. [76] did research utilizing data from Asian nations and discovered that EDU may not have a positive impact on ENQ in this specific area.

2.4. Globalization and Environmental Quality Nexus

GLN has stimulated economic expansion, exerting a substantial influence on the political, environmental, and economical aspects of human existence [6,77]. GLN is the term used to describe the growing interconnection and interdependence of governments, economies, and people on a global scale. GLN is defined as the rapid movement of commodities,

services, money, and information across national boundaries, as well as the growing interconnectedness of economies and society [78]. GLN is categorized into three distinct aspects in the literature: economic GLN, social GLN, and political GLN [79]. Several studies have focused on the impact of GLN on environmental degradation, although there is no widespread agreement [80]. All forms of GLN have a direct impact on the environment. According to the endogenous growth hypothesis, economic GLN is believed to assist nations in achieving sustainable long-term economic development [81]. Nevertheless, the effectiveness of environmental legislation in shaping ENQ depends on its design, enforcement, and industry response [82]. If policies are well-structured and strictly enforced, then they can drive significant environmental improvements. However, if regulations are poorly implemented or contain loopholes, then they may unintentionally lead to further degradation. For instance, some industries might adopt cost-cutting measures that bypass environmental safeguards, thereby undermining the intended benefits. Meanwhile, social GLN plays a crucial role in spreading awareness and promoting best practices, so that businesses can adhere to the highest sustainability standards.

Moreover, social GLN expedites the dissemination of information, specifically about the highest quality standards and protocols for doing business. Furthermore, GLN has an influence on ENQ via three distinct impacts: size, method, and composition. GLN's methodology and structure promote the adoption of eco-friendly technology and machine alterations, resulting in a reduction of negative environmental impacts [83]. GLN sometimes fosters the expansion of businesses that generate significant pollution, particularly in developing countries with lenient environmental restrictions [84]. The expansion of industrialization poses a threat to ENQ due to the increase in GHG emissions and the subsequent impact on global warming. Various criteria have been used in the literature to analyze the impact of GLN on the environment. Ref. [85] investigate the impact of GLN on the environment. It was shown that trade has a significant impact on the environment in terms of magnitude, even when considering the influence of composition and method. [78] used an ARDL model to analyze the impact of economic GLN and technology breakthroughs on environmental deterioration in South Asian countries from 1975 to 2017. The researchers used FDI and the KOF economic GLN index as indicators of GLN. The researchers examined a link between GLN, technological progress, and environmental degradation, finding evidence of an inverted U-shaped pattern in the sample countries. A study conducted by [6] investigated the association between GLN and EFT in Malaysia from 1971 to 2014. The findings indicated that although GLN has a limited impact on the EFT, it significantly increases the ecological carbon footprint. Ashraf [86] highlighted the importance of GLN and modernization, promoting the development of bilateral cooperation and strategic linkage between nations. Utilizing the FDI, they contended that GLN led to a rise in the EFT across the 75 nations included in BRI from 1984 to 2019. Ref. [87] used dynamic heterogeneous panel estimating methodologies to show that a 1% rise in economic GLN leads to a 0.11% reduction in CO₂ in 15 countries from 1970 to 2012. However, ref. [88] discovered that there was no impact of political GLN on the EFT in 146 countries. However, it was shown that although social GLN reduces the EFT, economic GLN increases it. Ref. [89] investigated the association between financial growth, GLN, and CO₂ emissions in the economies of the APEC from 1990 to 2016. The researchers used the Cup-BC and Cup-FM approaches to analyze the data. They found that the ENQ of APEC nations increased throughout this period because of these variables. Ref. [90] concurred that economic GLN also reduces pollution in the GCC countries when the research used a CS-ARDL estimator.

2.5. Natural Resources and Environmental Quality Nexus

Ecological degradation and the NTS nexus are crucial components of SSD. Whereas EFT assesses the degree to which human activities influence the environment, including the use of NTS and elements required to produce commodities and services necessary for human existence and well-being. Therefore, human activities are the primary catalysts for the exhaustion of NTS and the deterioration of the environment. Therefore, EFT may be reduced by optimizing the use of NTS and embracing more sustainable practices in manufacturing and usage activities. This can be done by adopting several actions such as the use of green energy, practicing recycling of materials, and implementing sustainable land use techniques. Moreover, it is important to know that NTS is used in a way that is both environmentally sustainable and fair for everyone on land. This involves implementing policies that advance the preservation and responsible use of NTS, safeguard biodiversity, and uphold the rights of native communities reliant on these resources. In simple words, NTS refers to the payments made for the use of resources. According to a study of [25], a rise in fees paid for the use of NTS has a beneficial impact on EFT. Therefore, the use of NTS is closely linked to human activities. Pollution occurs when contaminants are released into the environment. In past studies, we see that scholars have shifted their attention to the use of NTS. Hence, in the field of literature, NTS is often favored as a representative measure for quantifying natural resource use. The literature presents diverse outcomes on the influence of NTS on EFT. For instance, ref. [91] saw a positive correlation between these variables in their research conducted in Pakistan, but [92] discovered a negative correlation in the European Union. Furthermore, ref. [93] conducted a study to analyze the influence of NTS on environmental deterioration in SAARC nations. Their study used yearly data from 1996 to 2018. Their findings indicate that the environmental degradation in the SAARC is worsened by the depletion of NTS. Moreover, ref. [94] examined the impact of NTS on EFT in the most economically prosperous nations from 1990 to 2018. They analyzed that NTS has a positive impact on EFT in the long run and a one-way relationship from NTS to EFT was discovered. Similarly, ref. [33] investigate the effects of NTS on ENQ in 36 OECD nations. In their study they used yearly data from 2000 to 2018 and AMG and GMM methodologies. They found a positive correlation between NTS and CO₂. Furthermore, ref. [95] also examined the correlation between NTS and CO₂ in four ASEAN countries from 1990 to 2019. They used the CCEMG and AMG methodologies for their analysis. Their study showed that NTS effectively decreases CO₂. Moreover, ref. [96,97] investigated the correlation between NTS and ENQ in BRICS nations from 1990 to 2019. Conclusively, the literature shows that it was determined that NTS effectively decreases pollutants. Proceeding with the literature we see that [98] conducted a study on the influence of NTS on environmental pollution in France from 1990 to 2018. Their empirical analysis indicates that NTS has a considerable and detrimental impact on environmental deterioration. Further, digging deeper into the literature, ref. [99] discovered that NTS in E-7 nations had a positive effect on EFT based on yearly data from 1992 to 2020. Also, ref. [100] conducted a study to examine the impact of NTS on CO₂ in 93 countries. They used the AMG and MG methodologies, using yearly data from 1995 to 2017. Ref. [101] also examined the influence of NTS on CO₂ via the use of Quantile Regression, FMOLS, and DOLS methodologies. The researchers used yearly data for the MENA area spanning from 1990 to 2018. The research determined that the influence of NTS on CO₂ is negligible at the lower quantiles but becomes positive at the higher quantiles.

2.6. Urbanization and Environmental Quality Nexus

Numerous studies have explored the relationship between urbanization (URZ) and environmental quality, yet findings remain inconclusive. Some scholars argue that URZ

contributes to environmental degradation by increasing energy demand and emissions. For instance, ref. [19] demonstrated that URZ positively influences emissions in 23 European countries using the FMOLS approach. Similarly, panel data studies have linked URZ with rising energy consumption and emissions [102,103]. However, ref. [104] suggests that the impact of URZ varies across nations, with some experiencing a rise in CO₂ emissions while others witnessing a decline. Ref. [105] further documents the mixed effects of URZ on CO₂ emissions in South and Southeast Asian countries, showing that URZ exacerbates environmental degradation in medium- and high-income nations but has an insignificant impact in low-income countries. In contrast, ref. [106], using the ARDL technique, found that URZ negatively affects CO₂ emissions in Pakistan. Meanwhile, ref. [107] reported an insignificant link between URZ and emissions in 16 emerging nations, attributing variations to different estimation methods. Provincial-level research by [108,109] highlights disparities within China, revealing that URZ reduces energy consumption in lower-income provinces while increasing CO₂ emissions more significantly in middle-income regions. Additionally, ref. [110] employed panel data from 73 nations and concluded that the effects of URZ depend on development levels and urbanization intensity.

On the other hand, several researchers have examined URZ as a determinant of the ecological footprint (EFT) rather than emissions. Ref. [111] found that energy consumption, URZ, and trade openness collectively drive EFT in 14 MENA countries. Similarly, ref. [17] reported a negative association between URZ and environmental quality (ENQ), attributing this to economies of scale, service sector expansion, adoption of green technologies, and improved waste management in urban areas. Additionally, research suggests that URZ follows a nonlinear relationship with emissions. Ref. [112] found an inverted U-shaped correlation in developing nations, a pattern corroborated in Malaysia [113], Indonesia [40], and a broader sample of 144 countries [114]. However, refs. [115,116] failed to confirm this relationship. Given the complexity of URZ's impact, understanding causal dynamics is essential for effective policymaking. Prior studies indicate bidirectional [117,118], and even non-existent [106,119] causality between URZ and emissions. While reducing urbanization levels could theoretically lower emissions, a more viable approach involves promoting energy efficiency, environmental awareness, and green technologies. Additionally, examining the causal links between URZ, income, and energy consumption remains critical for formulating sustainable policies.

2.7. Research Gap

A review of the literature reveals that most studies have focused on examining the relationship between financial development, GDP, trade openness, energy consumption, GLN, URZ, and environmental quality. However, in recent years, researchers have increasingly explored the link between technological advancements and environmental sustainability. While some studies highlight TLI as a key strategy for mitigating environmental degradation and enhancing ENQ [120], others argue that the misuse of technology for industrial expansion can worsen ecological deterioration [121]. Despite the growing body of research on the environmental consequences of TLI, HMC, EDU, NTS, and GLN, no study has comprehensively examined their combined impact on the environmental quality of G20 countries. To address this gap, the present study employs the DKSE technique and CS-ARDL approach, which has not yet been explored in this context. The selection of G20 countries is justified by their crucial role in global environmental policymaking and the need to assess the effectiveness of key predictors in driving sustainability. Additionally, due to the high degree of cross-border information and technology exchange among these nations, analyzing the combined impact of GLN and TLI is essential. Such developments could foster advanced, eco-friendly technologies that align with sustainability goals. More-

over, literature often overlooks the heterogeneous effects of carbon emissions and ecological footprints across different economies, leading to potential biases. To bridge these gaps, this study uniquely applies to the DKSE technique to investigate the impact of technology footprints, HMC, EDU, NTS, GLN, and URZ in G20 countries from 2000 to 2022, while addressing critical research questions on sustainability and economic growth.

3. Methodology

This section provides an analysis of the variables, including their corresponding symbols and the origins of the data. This section presents the ongoing debate on econometric models and techniques.

3.1. Data Collection & Study Setting

This study examines the relationships between technological innovation (TLI), education (EDU), human capital (HMC), natural resources (NTS), globalization (GLN), and urbanization (URZ) with the ecological footprint (EFT) within the framework of Sustainable Development Goals (SDGs) in G20 nations from 2000 to 2022. The selection of G20 countries is justified by their substantial economic, political, and technological influence on global sustainability. Collectively, G20 nations account for approximately 85% of the world's GDP, 75% of global trade, and 60% of the global population, making them key drivers of environmental change [122]. Their substantial investments in research and development (R&D) further underscore their role in shaping technological innovation (TLI) and its environmental implications. Notably, developed G20 economies such as the United States, Germany, and Japan lead in education and human capital development, while emerging economies like Brazil, India, and China face distinct sustainability challenges, including resource overconsumption and rapid urbanization [123]. Additionally, resource-rich nations like Russia, Saudi Arabia, Brazil, and Indonesia are significant contributors to global natural resource exploitation, necessitating an investigation into the sustainability efforts of both developed and developing G20 countries [124]. Moreover, many G20 countries have committed to achieving SDGs by implementing policies aimed at renewable energy adoption, sustainable urban planning, and environmental governance. By analyzing the interplay between these factors and ecological footprint, this study aims to provide nuanced insights into sustainability disparities between developed and developing G20 nations and offer targeted policy recommendations for enhancing environmental sustainability across diverse economic contexts.

3.2. Econometric Model

This study adopts the paradigm advocated by previous studies on the topic of EFT, as shown by the works of [125,126]. Therefore, based on prior research, the model architecture for the present study has been specified in Equation (1).

$$EFT_{it} = \beta_0 + \beta_1 TLI_{it} + \beta_2 HMC_{it} + \beta_3 EDU_{it} + \beta_4 GLN_{it} + \beta_5 URZ_{it} + \beta_6 NTS_{it} + \varepsilon_t \quad (1)$$

Prior study has shown that converting data into its logarithmic form in panel data analysis is a widely used approach with several advantages. Applying variance stabilization techniques enhances the stability of the variance and increases the suitability of the data for linear modeling, such as fixed or random effects models [127]. Thus, this study converted the indicators into their logarithmic form, as represented by Equation (2).

$$\ln EFT_{it} = \beta_0 + \beta_1 \ln TLI_{it} + \beta_2 \ln HMC_{it} + \beta_3 \ln EDU_{it} + \beta_4 \ln GLN_{it} + \beta_5 \ln URZ_{it} + \beta_6 \ln NTS_{it} + \varepsilon_t \quad (2)$$

EFT is for ecological footprint, TLI represents technical innovation, GLN represents globalization, HMC stands for human capital, NTS represents natural resources, EDU represents education, and URZ shows urbanization. The model's intercept is represented as β_0 , whereas $\beta_1 - \beta_6$ reflect the coefficients of the Regressions in the study. The error term of the models is represented by ϵ_t , and $\ln$ denotes the logarithmic form for all the indicators.

3.3. Econometric Approaches

Before applying for the regression test, we performed several pre-tests to validate the data, model, and relationships among the variables. In the first stage, PCD was examined, followed by SLT, second-generation CIPS and CADF unit root tests, and the WLCT test. After these tests, we examined the relationships among the variables through DKSE, and their robustness was verified through the CS-ARDL method. To account for potential endogeneity arising from omitted variable bias, measurement errors, or reverse causality, we employ the CS-ARDL model, which helps mitigate these concerns by addressing cross-sectional dependence and incorporating lag structures. While instrumental variable (IV) techniques were considered, we relied on the model's econometric advantages in handling endogeneity within panel settings. However, future studies may explore IV-based approaches to further validate causal relationships. Finally, the panel causality test was used to examine the directional relationships among the study variables. The flowchart of the study is shown in Figure 3.

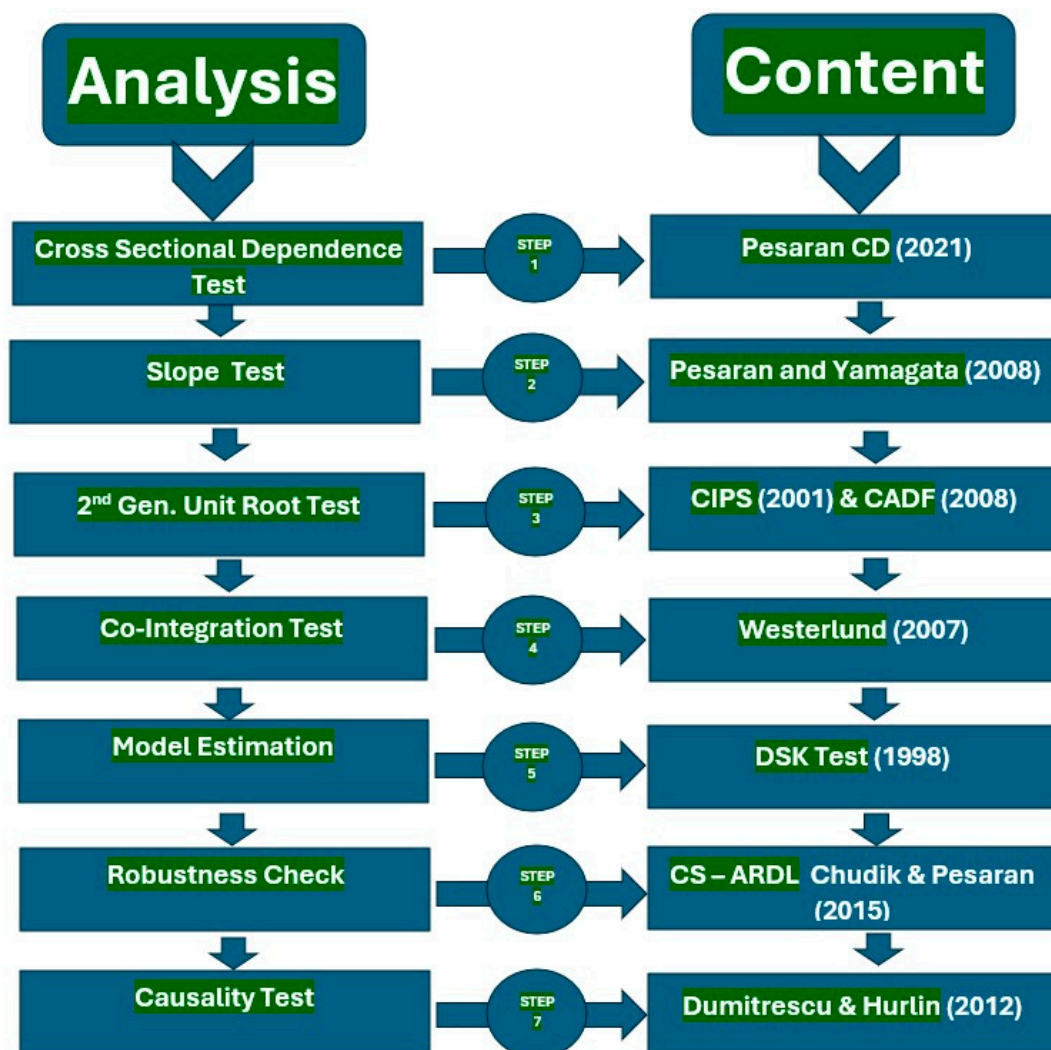


Figure 3. Study flow chart [128–131].

3.3.1. Pesaran Cross-Sectional Dependence (PCD)

Before moving towards the results of the regression test it is required to check for PCD in the dataset. PCD is considered important in the analysis of panel data, which encompasses observations throughout time and various cross-sectional units. The purpose of these tests is to identify and consider any interrelationships or associations between distinct cross-sectional units that may challenge the assumption of independence [132]. The reason for the occurrence of PCD might be due to geographical or regional spillovers, shared unobserved characteristics, or other impacts that affect the units during a certain period. In order to identify and resolve these relationships PCD is crucial for model estimation [133]. Hence, this research used three cross-sectional dependency tests, including the Breusch-Pagan LM test, the PCD test, and the Pagan LM coefficient approach. These tests were employed to ensure that the panel data analyses provide more dependable outcomes. The equations for the PCD proposed by [128] are represented by Equation (3).

$$PCD = \sqrt{\frac{2T}{N(N-1)}} \left(\sum_{i=1}^{n-1} \sum_{j=i+1}^n \sigma_{ij}^t \right) \quad (3)$$

N represents the dimension of PCD in the model. T also represents the temporal features of the data, and the calculated deviations of the indicators are shown using σ_{ij}^t . This test captures regimes and changes that indicate structural fractures, as shown by [134].

3.3.2. Slope Homogeneity Test

After checking the PCD the next pretest is checking the SLT. This test is crucial in assessing whether the correlation between the independent and dependent variables is consistent across different groups. Therefore, the SLT is used to detect variations in the data, suggesting that the impact of a certain variable may change across various entities or over different time periods [135]. Based on this test, the research may make more knowledgeable selections about the suitable model parameters and consider possible causes of variance in their panel data analysis, eventually resulting in more precise and significant outcomes. Hence, this work used the [129] test to investigate the SLT and the equations presented in Equations (4) and (5).

$$\tilde{\Delta}SLT = (N)^{\frac{1}{2}} (2K)^{-\frac{1}{2}} \left(\frac{1}{N} \tilde{S} - K \right) \quad (4)$$

$$\tilde{\Delta}ASLT = (N)^{\frac{1}{2}} \left(\left(\frac{2k(T-k-1)}{T+1} \right)^{-\frac{1}{2}} \right) \left(\frac{1}{N} \tilde{S} - K \right) \quad (5)$$

The symbol $\tilde{\Delta}SLT$ represents the delta slope, whereas $\tilde{\Delta}ASLT$ represents the adjusted delta SLT.

3.3.3. Unit Root Tests

Before applying [130] we used a second-generation unit root tests known as CIPS and CADF. These tests are essential statistical procedures in panel data analysis used to evaluate the stationarity of time series data. The design of these tests is based on the principle that analyzed variables show a unit root, which indicates non-stationarity [136]. Both CIPS and CADF tests are valuable for analyzing the presence of a unit root while accounting for the possibility of PCD [137]. These tests are considered important for ensuring reliable results in panel data analysis by accessing data stability and guiding the selection of suitable

econometric models [138]. The equations for the CIPS and CADF functions are given in Equation (6a) and (6b) respectively.

$$CADF = \gamma x_{it} = \alpha_{it} + \beta_{it-1} + \delta_I T + \sum_{j=1}^N \gamma_{ij} \gamma x_{it-j} + \varepsilon_{it} \quad (6a)$$

$$CIPS = N^{-1} \sum_{i=1}^N t_i(N, T) \quad (6b)$$

γ denotes the differences or inequalities among the indicators x_{it} , which are the variables being evaluated in this study.

3.3.4. Co-Integration Test

In panel data analysis, ref. [130] proposed a crucial co-integration test. This test is considered important in establishing long-term associations between variables in a panel dataset. The test considers the possible existence of different intercepts and slopes among various groups in the dataset. Furthermore, the [130] test has emerged as a crucial instrument for examining economic and social phenomena that extend across time and various groups, hence improving the comprehension of intricate interconnections within a panel data framework [139]. The equations for this co-integration test are shown in Equations (7) to (10).

$$G_\tau = \frac{1}{N} \sum_{i=1}^N \frac{\eta_i}{S.E(\hat{\eta}_i)} \quad (7)$$

$$G_a = \frac{1}{N} \sum_{i=1}^N \frac{T\eta_i}{1 - \sum_{j=1}^k \hat{\eta}_{ij}} \quad (8)$$

$$P_\tau = \frac{\hat{\eta}_i}{S.E(\hat{\eta}_i)} \quad (9)$$

$$P_a = T\eta_i \quad (10)$$

The group statistics mean is represented by $(G_t - G_a)$, whereas co-integration is denoted by $(P_t - P_a)$.

3.3.5. Regression Estimation Through DKSE

The current study uses the [140] estimation methodology to examine the relationships among the parameters. This test is useful for estimating the standard errors of regression coefficients in panel data analysis. Whereas panel data sets are collections of observations on many things across different time periods [141]. Panel data is extensively used by academics in empirical studies across numerous domains, including economics, finance, and social sciences. Its significance cannot be overstated [142]. Nevertheless, because of the possible connection between data within the same entity over time, using typical OLS standard errors may result in parameter estimates that are biased and inefficient. Whereas, DKSE solves this problem by using advanced statistical techniques to calculate more precise and resilient standard errors, considering the likely presence of heteroscedasticity and autocorrelation in panel data [143]. Moreover, the DKSE technique can significantly address the possible correlation in the panel data, resulting in more accurate estimations of the standard errors of the coefficients. Additionally, DKSE's technique enables researchers to draw more precise conclusions regarding the significance of their calculated coefficients, resulting in more dependable and resilient study outcomes. In addition, DKSE standard errors offer a practical solution to the issues commonly encountered when estimating standard errors in panel data analysis, particularly in relation to potential heteroscedasticity and autocorrelation. This makes DKSE standard errors an indispensable tool for applied econometric research [144]. More importantly, the DKSE estimate method tackles the issue of endogeneity by offering resilient standard errors that remain reliable even when

there is geographical, temporal, and cross-sectional dependency in the error terms. This approach accounts for potential correlation and heteroscedasticity without necessitating any alterations to the model or the use of instrumental variables. Based on the previous explanation, the study calculated the DKSE coefficients using the OLS approach, as shown in Equation (11).

$$y_{it} = x_{it-1}\beta + z_{it}y + \mu_{it} \quad i = 1, \dots, N, t = 1, \dots, T \quad (11)$$

The dependent variable EFT is represented by y_{it} , whereas the independent parameters (TLI, EDU, HMC, GLN, URZ, HMC) are indicated by x . Furthermore, the symbol μ represents the error term of the function. Furthermore, I depict the G20 countries and denote the specific time for the study, which spans from 2020–2022.

3.3.6. Robustness Test

Using the CS-ARDL estimation methodology in empirical research may serve as a reliable test to verify the accuracy of the DKSE estimation method. The paper can improve the reliability of the empirical findings by using CS-ARDL analysis. CS-ARDL has the advantage of modeling both time-series and cross-sectional dynamics, considering lagged relationships, and accommodating fixed effects. This analysis technique has been used by [145,146]. Utilizing this supplementary analysis may enhance the credibility of findings derived from the DKSE technique and provide a deeper understanding of the intricate relationship among parameters in panel datasets while accounting for unobserved differences and changes over time. The equation for the CS-ARDL model is explicitly defined in Equation (12).

$$\Delta \ln EFT_{i,t} = \delta_i + \sum_{j=1}^m \delta_{it} \ln EFT_{i,t-j} + \sum_{j=0}^m \delta_{it} X_{i,t-j} + \sum_{j=0}^1 \delta_{it} \bar{Z}_{i,t-j} + \mu_{it} \quad (12)$$

The symbol Z_t represents the PCD averages, denoted by $\Delta \ln EFT_{i,t}$ and X_t' . The variables X_{it} represent the explained variables in the research, namely TLI, EDU, HMC, GLN, URZ and NTS.

3.3.7. Panel Causality Test

Performing a regression analysis where the dependent variable (EFT) is regressed on other independent factors does not always imply causality, as stated by [147]. Hence, it is crucial to deduce the causal relationship between the variables under consideration. Following this approach, we use the [131] test to determine the causal connection between the EFT and TLI, EDU, HMC, GLN, URZ, and NTS. The BIC method is employed to identify the most suitable lag. The Dumitrescu and Hurlin causality test is an expanded version of the Granger causality test (1969) that is designed to analyze heterogeneous panel data with fixed coefficients for all units. We use this method because it considers the cross-sectional dependency across nations using a bootstrap operation that adjusts the critical values of the panel Granger causality test [131]. This test implies that there is no causal link between two variables, which is referred to as the null hypothesis (H_0).

4. Results

4.1. Descriptive Assessment of Study Variables

Table 1 presents the descriptive statistics for the studied indicators, highlighting variations in data consistency. Education (EDU) and Human Capital (HMC) exhibit relatively low standard deviations of 1.044316 and 0.568146, respectively, indicating minimal variability and a stable distribution across the selected countries. In contrast, Natural Resources (NTS: 9.150381), Globalization (GLN: 10.02458), and Urbanization (URZ: 15.24809) show the highest variability, reflecting substantial cross-country differ-

ences in resource availability, economic integration, and urban expansion. The skewness values suggest that most indicators are symmetrically distributed, except for NTS, which is highly positively skewed (3.4589), and URZ, which is negatively skewed (−1.3839), indicating an uneven spread of values. Additionally, NTS exhibits a leptokurtic distribution (15.68916), characterized by a sharp peak and heavy tails, while HMC has a platykurtic distribution (1.735647), suggesting a flatter distribution. The negative skewness observed in Education (EDU: −0.1331) and Human Capital (HMC: −0.1328) implies that their values are slightly concentrated at the higher end of the scale. These variations may stem from differences in national policies, economic structures, urbanization trends, and the fact that the data were collected from multiple sources, each with its own methodologies and classifications. Such factors influence the distribution of these indicators, leading to variations in consistency and dispersion. The findings highlight the diverse economic and environmental conditions across the selected countries, further contributing to differences in data patterns.

Table 1. Descriptive statistics.

	EFT	EDU	HMC	GLN	NTS	TLI	URZ
Mean	4.281711	4.450140	2.967308	72.04082	4.729452	10.83145	72.94735
Median	4.220042	4.525985	2.974337	70.18541	1.920028	10.67926	78.38700
Maximum	10.92681	7.718870	3.842989	90.04584	55.47506	25.23000	92.34700
Minimum	0.087267	1.264700	1.782071	46.40732	0.010688	1.450000	27.66700
Standard Deviation	2.325017	1.044316	0.568146	10.02458	9.150381	3.675972	15.24809
Skewness	0.3247	−0.1331	−0.1328	0.1391	3.4589	0.8877	−1.3839
Kurtosis	2.936457	2.900538	1.735647	1.952404	15.68916	4.990979	4.262370

4.2. Outcomes of Cross-Sectional and Slope Test

Panel data analysis begins by examining the presence of PCD between the series using various tests. To provide accurate and consistent findings, it is necessary to use procedures that consider the presence of a PCD between series. Failure to do so may introduce bias and inconsistency in the results [148]. Hence, we examine the inter-series PCD using the test formulated by [128]. The outcomes of these estimations are shown in Table 2. The data shown in Table 2 provide evidence for the PCD between the series. All series are rejected at the 1% significance level. This indicates that any shock that happens in the sample nations also has an impact on the other countries within the G20 area.

Table 2. PCD test.

Test	Statistic	Prob.
Pagan LM	1554.096 ***	0.0000
Scaled LM	79.06635 ***	0.0000
Pesaran CD	21.26611 ***	0.0000

Note: *** $p < 1\%$.

Once the existence of PCD in the dataset has been confirmed, it is essential to assess the homogeneity or heterogeneity of the data's slope. The hypothesis of the SLT states that the variables are homogenous, meaning they have similar characteristics, and the nations being studied do not differ significantly. The findings of the SLT are shown in Table 3. The results suggest that the data exhibits heterogeneity, leading to the rejection of the null

hypothesis of SLT. The p -values for both delta and adj. delta are 0.000, showing a significant difference in the slope coefficients. This indicates that there is a wide range of variation and distinction across the G20 nations.

Table 3. Slope test.

Test	Value	Prob.
Delta	21.439 ***	0.000
Adj. Delta	23.973 ***	0.000

Note: *** $p < 1\%$.

4.3. Outcomes of CIPS and CADF

The results from the CIPS and CADF tests validate the stationarity characteristics of the variables. While both tests assess unit root properties, they differ in their methodological approach. The CIPS test accounts for cross-sectional dependence, making it more suitable for panel data with interdependence, whereas the CADF test extends the ADF framework to handle heterogeneity across cross-sections. As shown in Table 4, neither test confirms stationarity at level ($I(0)$); however, all variables become stationary at the first difference ($I(1)$). These findings ensure the reliability of the data for further econometric analysis, as stationarity at first difference prevents spurious regression issues.

Table 4. Unit root tests.

Variables	CIPS Test		CADF Test	
	I(0)	I(1)	I(0)	I(1)
EFT	−0.808	−10.297 ***	0.432	−8.145 ***
TLI	−1.589	−5.467 ***	−1.254	−11.247 ***
HMC	4.533	−6.073 ***	4.324	−7.514 ***
NTS	−0.466	−4.759 ***	−1.019	−6.017 ***
EDU	−1.342	−12.489 ***	−1.428	−7.661 ***
URZ	−1.264	−7.125 ***	−1.324	−13.547 ***
GLN	−2.317	−11.365 ***	−2.007	−15.257 ***

Note: *** $p < 1\%$ Where $I(0)$ and $I(1)$ show stationarity at level and at first difference respectively.

4.4. Outcome of Westerlund Co-Integration (WLCT)

To ensure the robustness of our analysis, we examine long-term co-integration among the study variables using the Westerlund co-integration test (WLCT), which is particularly suitable for datasets exhibiting panel cross-sectional dependence (PCD) and slope heterogeneity. The null hypothesis (H_0) of the WLCT states that no co-integration exists among the time series, while the alternative hypothesis (H_1) suggests the presence of long-term equilibrium relationships. The results, presented in Table 5, indicate that the test statistics Gt and Pt reject H_0 at the 1% significance level, confirming the presence of co-integration among the variables. However, Ga and Pa have p -values close to or equal to 1, indicating that these statistics fail to reject the null hypothesis. The observed variation in results stems from the methodological differences between the test statistics: Gt and Pt focus on testing co-integration at the individual panel level, while Ga and Pa assess co-integration across the entire panel. Since Gt and Pt are considered more robust in cases with cross-sectional dependence and heterogeneity, their statistical significance provides strong evidence of co-integration among the studied variables. Thus, despite the non-significant results for

Ga and Pa, the overall findings confirm the existence of a stable long-term equilibrium relationship in the dataset.

Table 5. Westerlund Co-integration test.

Statistics	Value	Z-Value	p-Value
Gt	−4.464 ***	−7.187	0.000
Ga	−8.913	3.890	1.000
Pt	−15.830 ***	−4.551	0.000
Pa	−9.442	1.827	0.966

Note: *** $p < 1\%$.

4.5. Results of DKSE

We investigate the long-run relationships of TLI, HMC, EDU, NTS, GLN, and URZ with EFT for G20 countries using DKSE from 2000 to 2022. Table 6 presents the results of the DKSE test, offering insights into the situations of G20 countries. Moreover, Figure 4 displays the graphical abstract of the study.

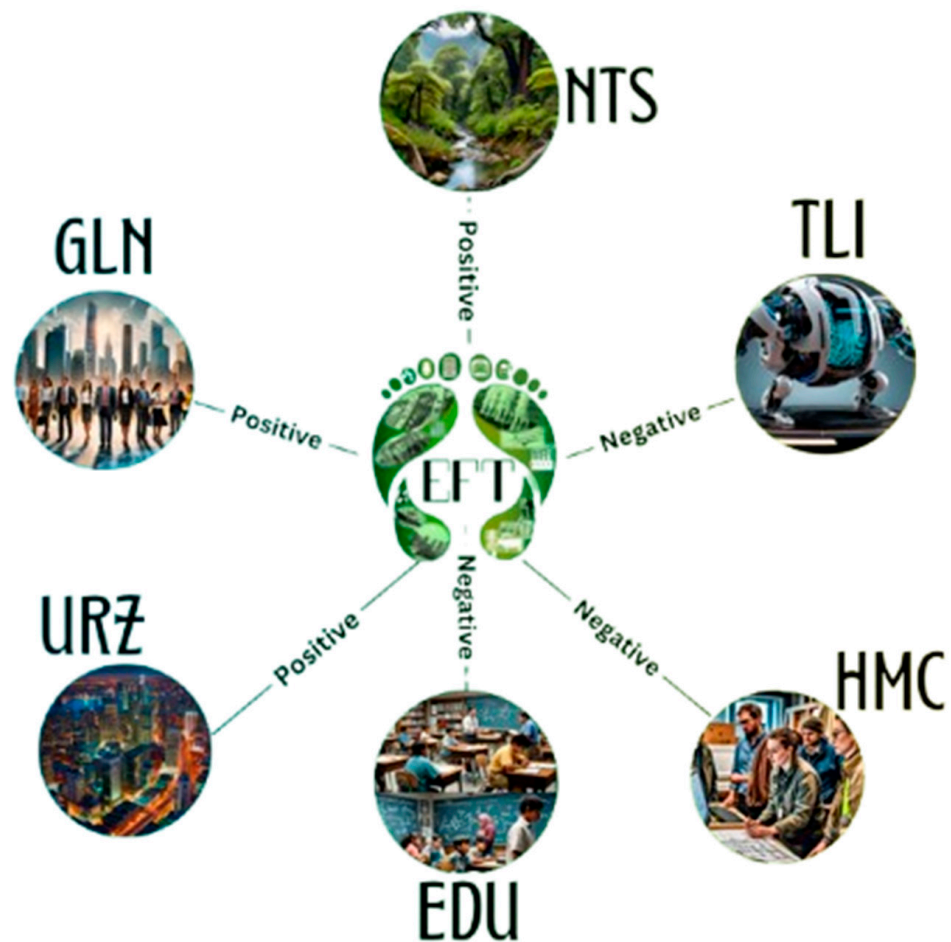


Figure 4. Study Outcomes.

Table 6. DKSE test.

Variables	Coefficient	Std. Error	p-Value
EDU	−0.231013 ***	0.040128	0.000
HMC	−4.012112 ***	0.712312	0.001
GLN	0.112321 ***	0.003772	0.004
NTS	0.123010 ***	0.013143	0.000
TLI	−0.040123 ***	0.032417	0.000
URZ	0.201323 ***	0.032123	0.002
Constant	12.837 ***	0.0497	0.000
Diagnostic test (<i>p</i> -value = 0.000)			
R2	0.954	Jarque Bera	5.238 (0.362)
Adj. R2	0.962	Wald	513.327 (0.000)
F-Statistics	37.26 ***	Wooldridge	267.108 (0.869)

Note: *** $p < 1\%$.

4.6. Discussions

The study investigates the long-run relationships of TLI, HMC, EDU, NTS, GLN, and URZ with EFT among the G20 bloc. We ran the analysis using the DKSE test, and discussed the empirical findings as follows: The parameter EDU has a negative relationship with EFT, and as a result, EDU improves the ENQ of the G20 bloc. The negative coefficient value of EDU (−0.231013) indicates that a 1% increase in EDU will decrease the EFT at the rate of 0.2310%. The negative association between EDU and EFT in G20 countries is particularly noteworthy, as EDU fosters environmental awareness and encourages sustainable practices at both individual and national levels. Educated individuals better understand the long-term consequences of human activities on the environment, leading to the adoption of greener behaviors and support for stricter environmental policies. G20 nations are actively enhancing their education systems by integrating sustainability-focused curricula, promoting renewable energy adoption, and advancing recycling initiatives. Additionally, EDU plays a crucial role in driving technological innovation through research and development (R&D), which significantly improves environmental quality (ENQ). Investments in education and innovation not only enhance ecological sustainability but also have substantial financial implications. By allocating resources toward green research and eco-friendly technologies, G20 countries stimulate economic growth, create sustainable job opportunities, and strengthen industrial competitiveness. These investments lead to long-term economic resilience by reducing reliance on environmentally harmful industries and fostering sustainable business models. Moreover, highly educated societies tend to advocate stringent environmental regulations, which further promote sustainable economic practices. The G20's commitment to TLI and green technologies, supported by education-driven R&D, highlights the economic benefits of investing in sustainability. Our findings align with previous studies [20,149,150], reinforcing the notion that education not only improves ENQ but also contributes to economic sustainability by driving innovation and regulatory advancements. The results validate the null hypothesis based on our estimation.

Furthermore, the empirical result from the DKSE estimation reveals an inverse relationship between HMC and EFT, with a coefficient value of −4.012112. It implies that a 1% increase in HMC will reduce the EFT by 4.012%; therefore, HMC serves as a beneficial parameter for improving the ENQ of G20 countries. HMC is considered an asset because it can influence any country's economic conditions, productivity, and development. HMC primarily encompasses a worker's level of EDU, their skill set, their physical and mental

health, their practical experience, their abilities and capacities to solve problems, their social and cultural values, and their level of motivation and adaptability. Therefore, HMC does not solely consist of EDU. EDU is also just one factor that contributes to HMC formation. That's why we used HMC as a separate variable in our study, along with an educational parameter, to examine its impact on the ENQ of G20 nations. Possible explanations for the inverse relationship between HMC and EFT could include the following: As EDU is a member of the HMC index, schooling can bring environmental awareness among the nations through empirical teaching and training methods and continuous work on R&D. Just as EDU positively impacts the ENQ of countries, HMC similarly contributes to positive changes in the environmental settings of G20 nations. Secondly, HMC promotes green technologies and renewable energies, which in turn have a positive impact on the ENQ of any nation. The G20 countries are supportive of green innovation and green energy, leading to improvements in their environmental settings through the invention and utilization of green technologies. This progress is made possible by the efficiency and effectiveness of HMC. Furthermore, we see that G20 countries are transitioning from production to service industries. The literature shows that service industries place less burden on environmental pollution as compared to manufacturing industries. Therefore, these types of shifts are leading to improvements in the ENQ of G20 nations. The results of our study align with those of [21,151]. Furthermore, our study findings support the null hypothesis in our estimations.

In G20 countries, the study findings revealed a positive association between GLN and EFT with a coefficient value of 0.112321. This positive association suggests that a 1% increase in GLN leads to a 0.1123% increase in EFT. GLN is a critical factor that has a significant impact on any country's ENQ. GLN promotes the integration of economies, cultures, and societies through trade, investment, technology, and communication. The positive correlation between GLN and EFT in G20 countries may stem from various factors. For instance, GLB facilitates international trade, which increases economic growth and in turn, increases human activities, leading to an increase in EFT. G20 countries are the leading players in global trade. Due to GLN, their trade activities increase production levels, which in turn leads to the extraction of resources, the consumption of energy, and the generation of waste. Such activities lead to an increase in environmental pollution and a rise in EFT, thereby negatively impacting the ENQ of G20 countries. Moreover, economic growth driven by globalization (GLN) often accelerates urban expansion, which subsequently intensifies resource consumption and contributes to higher ecological footprints (EFT), thereby degrading environmental quality (ENQ) in G20 countries. This relationship stems from increased infrastructure development, energy demand, and industrial activities accompanying urbanization. Additionally, globalization fosters expanded trade and international travel, leading to higher freight transport and aviation activities. These factors significantly elevate greenhouse gas (GHG) emissions, further amplifying the ecological footprint. The interconnected nature of these variables underscores the complex link between globalization, urbanization, and environmental sustainability. As a result of these activities, the ENQ of the region deteriorates. Our findings align with the results of studies carried out by [152,153]. The results also support the null hypothesis of our study.

Moreover, the study explored the impact of NTS on EFT, and the results from the DKSE test show that NTS has a positive association with EFT. The direct relationship between NTS and EFT has a coefficient value of 0.123010. This finding indicates that a 1% increase in NTS is associated with a 0.1230% rise in EFT, suggesting that the exploitation of natural resources is driving environmental degradation in G20 countries. This implies that natural resources are being consumed at a rate that exceeds their natural replenishment, leading to sustainability concerns. The results highlight the need for balanced resource management to mitigate the adverse environmental impact. Several factors contribute to the positive

correlation between NTS and EFT in G20 countries. For instance, deforestation is one of the causes. Plants and trees in nature have a number of benefits for humans, animals, birds, and the environment as a whole. Cutting down these trees reduces biodiversity, increases CO₂ emissions, increases EFT, and negatively impacts the quality of the environment. In the past, Brazil has experienced deforestation in the Amazon rainforest due to logging, cattle ranching, and farming. Furthermore, Canada's 20th-century tar sand project led to large-scale deforestation and numerous environmental challenges. Another reason for the positive connection of NTS with EFT is coal mining for energy requirements. For example, in the G20, China and India are the leading countries that heavily rely on coal mining for energy consumption, leading to CO₂ emissions, an increase in EFT, and the spread of various health issues. The extraction of NTS in the region primarily causes these factors to escalate the level of environmental pollution. Hence the rapid increase in the extraction of NTS in the G20 can have negative impacts on the ENQ; therefore, G20 countries should restrict the extraction of NTS in the region to control EFT for the betterment of ENQ. Our results align with the findings of [94,99,154] and support the null hypothesis based on our estimates.

Moving on to the results, we find that TLI has a negative association with EFT in G20 countries, with a coefficient value of -0.040123 . This indicates that a 1% increase in TLI leads to a 0.0401% reduction in EFT. TLI involves the creation of new technologies and the enhancement of existing ones, improving processes, products, and services. It directly influences economic growth, SSD, climate change mitigation, urban development, e-commerce, and security systems. G20 countries leverage TLI to maintain their global competitiveness, with advanced technologies enhancing industrial efficiency, waste recycling, and pollution control, thereby promoting ecological balance. Furthermore, specific green technologies play a crucial role in reducing EFT. Renewable energy technologies, such as solar panels, wind turbines, and smart grids, contribute to cleaner energy production and lower carbon emissions. Carbon capture and storage technologies help mitigate industrial pollution, while advancements in energy-efficient manufacturing and green supply chain management optimize resource use. Additionally, smart transportation systems, electric vehicles (EVs), and green building innovations further support ecological sustainability by reducing reliance on fossil fuels and minimizing urban pollution. Additionally, TLI indirectly affects EFT by strengthening HMC. As technological advancements reshape industries, they elevate educational standards, workforce skills, and research capabilities. This, in turn, fosters greater awareness and adoption of environmentally friendly practices, reinforcing sustainability efforts. The significant R&D investments made by G20 nations further drive both technological progress and human capital development, amplifying the long-term environmental benefits of innovation. These interactions contribute to the observed negative association between TLI and EFT. Our findings align with prior research [155,156], supporting the study's null hypothesis and underscoring the importance of increased investment in TLI—particularly in green technologies—to enhance ENQ.

Finally, the results of this study show that URZ has a positive relationship with EFT in the G20, with a coefficient value of 0.201323 . At the 1% level of significance, the association between the variables is significant. The results indicate that a 1% increase in URZ will result in a corresponding 0.2013% increase in EFT. URZ is considered an economic powerhouse that drives economic growth by generating employment, expanding trade, fostering businesses, and promoting innovation. However, this economic expansion often comes at the cost of environmental quality, creating a fairness-accuracy trade-off in sustainability policies. While urban centers contribute significantly to national GDP through industrialization, tourism, and commercial activities, they also lead to increased resource consumption, waste generation, and air pollution, all of which elevate EFT. The

tourism industry, a key driver of urban economies, exemplifies this trade-off. While it boosts revenue and job creation, it also strains natural resources and leads to higher carbon emissions. Similarly, industrial activities in urban regions drive technological progress and economic prosperity but also contribute to air and water pollution, negatively affecting ENQ. Additionally, large-scale land clearance for urban expansion reduces biodiversity, exacerbating ecological degradation. While developed G20 nations such as the USA, UK, Canada, France, Germany, and Japan employ advanced green technologies to mitigate these adverse effects, developing nations like China, Argentina, Brazil, India, Russia, Saudi Arabia, Turkey, and Mexico face greater challenges due to the reliance on less efficient technologies, resulting in a more pronounced increase in EFT. This interplay between sustainability and economic growth underscores the importance of comprehensive urban planning and investment in green technologies. Striking a balance between economic expansion and environmental sustainability is crucial. While prioritizing sustainability may introduce short-term economic adjustments, long-term benefits—such as resource efficiency, lower healthcare costs due to reduced pollution, and enhanced climate resilience—can outweigh these costs. Therefore, implementing policies that integrate sustainable urbanization with economic growth is essential for mitigating the environmental impact of URZ in G20 countries. Our study's findings align with those of [17,157,158], and this outcome also validates our estimation's null hypothesis.

The diagnostic test results explain the quality and reliability of the regression model. The value of R^2 (0.954) shows the proportion of variance in EFT that is predictable from independent variables TLI, EDU, HMC, GLN, URZ, and NTS. The R^2 value reveals that the model explains 95.4% of the variability in the dependent variable, demonstrating the model's reliability with the data. Furthermore, the R^2 value of 0.962 modifies the R^2 values for TLI, EDU, HMC, NTS, GLN, and URZ within the model. This provides a more accurate measure when multiple independent variables are used. The high value of 0.962 indicates that the model's explanatory power is very high. Furthermore, the F-statistics value (37.26) with a p value less than 0.01 indicates the overall significance of the regression model. This value indicates the model used in the study has a high level of significance. Moreover, the Jarque-Bera test (5.238) with a p value of 0.362 suggests that residuals are not significantly different from a normal distribution according to the DKSE test. Finally, the values of the Wald test (513.327) and Wooldridge test (267.108) show that the model's predictors are significant and there is no significant autocorrelation in the residuals, respectively. Overall, these results show that the regression model is robust and reliable in estimations.

4.7. Robustness Results of CS-ARDL

We conducted a robustness check using the advanced CS-ARDL model, which provides both short-run and long-run estimates, as presented in Table 7. The results confirm the consistency of our findings with the DKSE model. In both the short and long run, TLI, EDU, and HMC exhibit a negative relationship with EFT, while GLN, URZ, and NTS show a positive association with EFT in G20 countries from 2000 to 2022. The negative and significant error correction term (ECT) confirms the model's stability and long-term adjustment process. Regarding the significance levels, we do not selectively assign p -values but rather report them based on the statistical results obtained. In the long run, all variables are significant at the 1% level ($p < 0.01$), indicating strong and consistent relationships over time. In the short run, variations in significant levels (1%, 5%, and 10%) reflect the different strengths of immediate effects across variables. This difference suggests that some variables exert stronger short-term fluctuations, while their long-term impacts stabilize and become more pronounced. The negative long-run relationships of TLI, EDU, and HMC with EFT imply that advancements in technology, education, and human capital contribute

to reducing environmental degradation over time. Conversely, the positive relationships of GLN, URZ, and NTS with EFT suggest that globalization, urban expansion, and natural resource exploitation drive environmental degradation. However, their short-run effects vary in magnitude, highlighting the dynamic nature of these factors in influencing ecological outcomes.

Table 7. CS ARDL test (Robustness check).

Variable	Coefficient	Std. Error	t-Statistic	Prob.
Long Run Equation				
EDU	−0.174225 ***	0.052388	−3.325627	0.0010
HMC	−3.953044 ***	0.600465	−6.583308	0.0000
GLN	0.035838 ***	0.001043	3.540919	0.0016
NTS	0.078196 ***	0.017612	4.439852	0.0000
TLI	−0.021745 ***	0.010151	−1.692896	0.0024
URZ	0.190878 ***	0.018345	10.40479	0.0000
Short Run Equation				
ECT	−0.305007 ***	0.071706	−4.253550	0.0000
EDU	−0.100889 **	0.052445	−1.645656	0.0174
HMC	−10.24520 **	8.199757	−1.235352	0.0291
GLN	0.022253 ***	0.011909	1.598768	0.0031
NTS	0.197765 **	0.236167	0.783008	0.0441
TLI	−0.035075 *	0.019515	−1.797339	0.0734
URZ	5.501025 *	3.045688	1.806168	0.0720

Note: *** $p < 1\%$, ** $p < 5\%$, * $p < 10\%$.

4.8. Outcomes of Dumitrescu Hurlin Causality Test (DHCT)

We have seen the interconnection and long-run association among the study variables TLI, EDU, HMC, GLN, URZ, NTS, and EFT. Now, it is required to see the direction of these relationships to understand how variables influence each other over time. Therefore, with the help of these causality relations, we can identify which variable is a cause and which variable is an effect in relationships. Causality relationships assist policymakers in making correct and effective decisions. Furthermore, the term causality refers to the idea that we use the past values of independent variables to predict the current values of dependent variables. This concept is based on a study by [159]. Table 8 presents the results of DHCT.

Based on the outcomes, it is evident that there is a dual causality or bi-directional causality between NTS, EDU, and EFT. Both NTS to EFT and EFT to NTS exhibit positive granger causality. The relationship between EDU and EFT is negatively correlated, whereas the relationship between EFT and EDU is positively correlated. A higher level of EDU may increase citizens' responsibility for ENQ, while an increase in EDU may lead to a decrease in EFT. Conversely, a positive correlation exists between EFT and EDU, suggesting that an increase in population leads to an increase in EFT. However, at primary and secondary EDU levels, individuals tend to overlook EFT, leading to a decline in ENQ. By controlling the population level and self-awareness through EDU at the primary and secondary levels, we can improve the ENQ in G20 countries. Furthermore, there is a unidirectional causal relationship between TLI and EFT. This implies that while TLI contributes to EFT, EFT does not contribute to TLI in the G20 countries between 2000 and 2022. Moving forward, we see that there exists uni-directional causality from URZ to EFT, HMC to EFT, and GLN

to EFT. However, the G20 countries do not exhibit any causality between EFT and URZ, EFT and HMC, or EFT and GLN. With the help of these relationships, policymakers can make effective and timely decisions to control and improve the G20 bloc's ENQ. Figure 5 displays the graphical representation of these results.

Table 8. Pairwise panel causality test.

Null Hypothesis	Z.Bar-Stat	p-Value	Decision
NTS $\nrightarrow$ EFT	3.351 ***	0.0000 ✓	Bi-directional
EFT $\nrightarrow$ NTS	5.042 ***	0.0000 ✓	
EDU $\nrightarrow$ EFT	−7.21333 ***	0.0000 ✓	Bi-directional
EFT $\nrightarrow$ EDU	6.10774 ***	0.0013 ✓	
TLI $\nrightarrow$ EFT	−4.48925 ***	7×10^{-6} ✓	Uni-directional
EFT $\nrightarrow$ TLI	1.13135	0.2579 ✗	
URZ $\nrightarrow$ EFT	7.21333 ***	0.0000 ✓	Uni-directional
EFT $\nrightarrow$ URZ	0.49419	0.6212 ✗	
HMC $\nrightarrow$ EFT	−3.00574 ***	0.0026 ✓	Uni-directional
EFT $\nrightarrow$ HMC	0.91883	0.3582 ✗	
GLN $\nrightarrow$ EFT	8.09049 ***	7×10^{-16} ✓	Uni-directional
EFT $\nrightarrow$ GLN	0.99758	0.3185 ✗	

Note: *** $p < 1\%$.

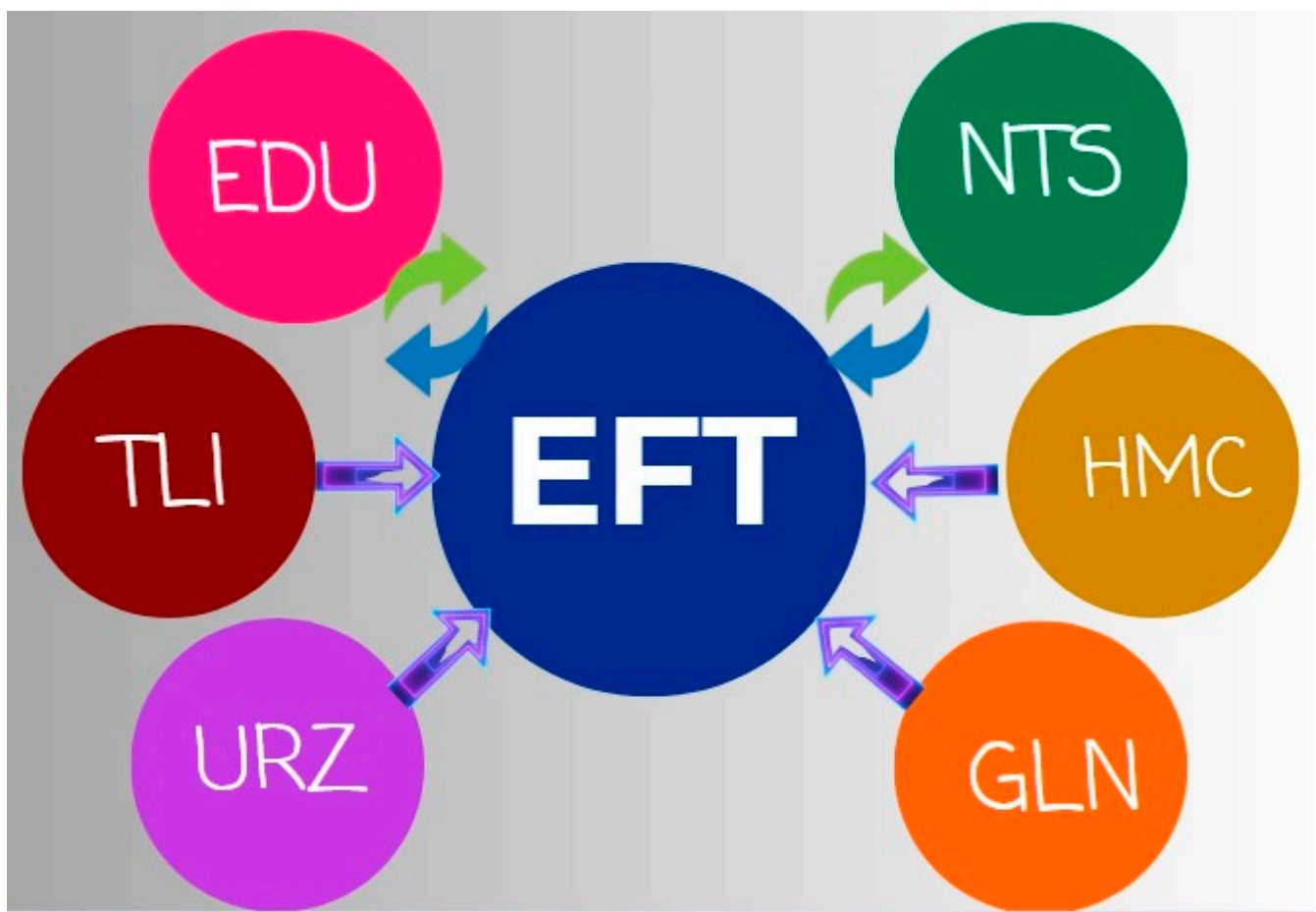


Figure 5. DHCT relationship outcomes.

5. Conclusions and Policy Implications

Governments worldwide are actively working to enhance economic and industrial systems in pursuit of environmental sustainability, aligning with the broader objectives of the SDGs. This study investigates the relationships between natural resources (NTS), technological innovation (TLI), governance (GLN), education (EDU), urbanization (URZ), and human capital (HMC) with ecological footprint (EFT) in G20 nations from 2000 to 2022. Utilizing the DKSE model and validating results through CS-ARDL, the findings reveal that TLI, EDU, and HMC contribute to reducing EFT, while NTS, GLN, and URZ exacerbate environmental pressure. Additionally, causality results suggest a bidirectional link between EDU, NTS, and EFT, while GLN, URZ, TLI, and HMC exhibit a unidirectional influence on EFT.

These findings underscore the importance of technology-driven sustainability, education-led environmental consciousness, and governance reforms in mitigating ecological degradation. Policymakers should focus on fostering green technological innovation, strengthening governance mechanisms, and regulating resource exploitation to achieve sustainability targets. Furthermore, integrating urban planning with environmental policies and promoting education-based sustainability awareness can help reduce ecological footprints. Given the G20's global economic influence, these strategies can set a precedent for achieving a balance between economic growth and environmental preservation.

5.1. Policy Recommendations

To ensure sustainable environmental quality in G20 countries, policymakers must implement targeted strategies that address the key drivers of ecological degradation. Given the varying impacts of technological innovation, globalization, urbanization, and natural resource exploitation on the ecological footprint, a balanced policy approach is essential. This requires not only integrating environmental considerations into economic and industrial frameworks but also establishing clear implementation pathways. To enhance the effectiveness of these strategies, G20 countries should strengthen coordination mechanisms through international environmental agreements, joint research initiatives, and technology-sharing platforms. Additionally, financing models such as green bonds, public-private partnerships, and climate funds should be leveraged to support the transfer and adoption of green technologies. By aligning policy actions with structured implementation frameworks—encompassing education reform, sustainable urban planning, and robust regulatory enforcement—governments can mitigate adverse environmental effects while fostering long-term sustainability. The following policy recommendations outline actionable steps to achieve these objectives effectively.

- As technological innovation (TLI) negatively impacts the ecological footprint (EFT), policymakers should promote eco-friendly technological advancements by investing in green research and innovation while encouraging industries to adopt cleaner production methods.
- Since globalization (GLN) contributes to rising EFT through increased trade and transportation, regulatory authorities must implement stricter environmental policies, such as carbon taxes and emission regulations, to reduce its adverse effects.
- As urbanization (URZ) intensifies environmental degradation, governments should integrate sustainable urban planning by enhancing public transportation, expanding green spaces, and promoting energy-efficient infrastructure.
- Given that natural resource exploitation (NTS) significantly increases EFT, policymakers should enforce responsible extraction policies, invest in renewable alternatives, and encourage circular economy practices to minimize environmental harm.

- Since education (EDU) and human capital (HMC) are associated with reduced EFT, authorities should integrate environmental awareness into educational curricula, provide incentives for green skill development, and promote sustainability-driven workforce training.
- To address the environmental impact of trade and transport, international organizations should mandate the adoption of cleaner technologies in shipping, aviation, and logistics while supporting carbon-neutral supply chains.
- As environmental challenges require global collaboration, G20 nations should strengthen international partnerships, share best practices, and implement unified sustainability policies to achieve long-term ecological balance.

5.2. Study Limitations & Recommendations

Although the current research thoroughly examines the influence of six significant components (TLI, EDU, HMC, GLN, URZ, and NTS) on EFT in G20 nations, it is important to acknowledge certain limitations. The study relies on data from 2000 to 2022, which, while comprehensive, represents a relatively short timeframe. Extending the period of analysis in future research could provide deeper insights into long-term environmental trends. Additionally, this study focuses on a limited set of environmental indicators, with EFT as the sole dependent variable. Future studies may incorporate broader ecological measures such as biodiversity loss, water stress, green growth, and load capacity to present a more holistic view of environmental quality (ENQ). Furthermore, this research primarily considers economic, technological, and urbanization-related factors but does not account for institutional quality, political stability, and governance, which could significantly influence environmental outcomes. Future research may integrate these aspects to provide a more nuanced understanding of sustainability dynamics. Expanding the scope beyond the G20 bloc to include other regional groupings such as APEC, BRICS, E7, G7, and CAREC could also enhance comparative insights. Additionally, emerging variables like artificial intelligence, digital commerce, biofuels, stock market capitalization, and the agricultural economy may further refine the analysis. Methodologically, this study employs CS-ARDL and DKSE, which, while robust, may be complemented by alternative econometric techniques in future research. Approaches such as MMQR, bootstrapping, Dynamic Common Correlated Effects, Panel Smooth Transition Regression, and Foreign Direct Investment (FDI) models could provide deeper insights into long-term relationships. By adopting these advanced methodologies and incorporating a broader set of influencing factors, policymakers can develop more innovative, adaptive, and effective strategies to mitigate EFT, achieve carbon neutrality, and fulfill the objectives of SDG-7, SDG-8, and SDG-13.

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Abbreviations

Abbreviation	Full form	Abbreviation	Full form
AIC	The Akaike Information Criterion	GLN	Globalization
BIC	Bayesian Information Criterion	HMC	Human Capital
CDT	Cross Sectional Dependence Test	NTS	Natural Resources
DHCT	Dumitrescu Hurlin Causality Test	PCD	Pesaran Cross Sectional Dependence
DKSE	Driscoll and Kraay Standard Error	PWT	Penn World Tables
EDU	Education	SLT	Slope Homogeneity
EFT	Ecological Footprints	TLI	Technological Innovation
EKC	Environmental Kuznets Curve	URZ	Urbanization
ENQ	Environmental Quality	WIPO	World Intellectual Property Organization
SSD	Sustainable Development	WLCT	Westerlund Co-integration

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